

Research paper

Development and optimization of a green hypergolic gel propellant based on gelled ethanol with aluminum powder and copper(II) chloride with 90% hydrogen peroxide as oxidizer

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ABSTRACT

Propellants based on hydrazine and its derivatives, combined with nitrogen tetroxide or nitric acid, are highly toxic, hazardous to handle, and harmful to the environment. Green propellants have been studied as replacements for these toxic propellants. One alternative is to use concentrated hydrogen peroxide as the oxidant and a gel propellant as the fuel. In this study, compositions of a hypergolic gel based on ethanol and Carbopol, containing aluminum particles and using copper(II) chloride dihydrate as the catalyst, were evaluated in terms of propulsive parameters, such as theoretical specific impulse, characteristic velocity, and volumetric specific impulse. The Rotatable Central Composite Design (RCCD) method was applied to optimize the fuel composition using 90% by weight hydrogen peroxide as the oxidizer. The propulsive parameters were estimated by using Chemical Equilibrium with Applications (NASA CEA) software. For the I_{sp} and c^* optimization, the composition which showed the highest values of both responses, 172.8 s and 1568 m/s, respectively, was 97.5% of gel based on 95% ethanol and 2.5% Carbopol, 2% copper chloride dihydrate, 0.5% aluminum particles. For $\rho_{I_{sp}}$ optimization, the composition which yielded the highest value of this response (212315 kg s/m³) was 89.5% of gel based on 87% ethanol and 2.5% Carbopol, 10% copper chloride dihydrate and 0.5% aluminum particles. Both optimizations had the highest value at an oxidizer-fuel ratio of 4. These findings demonstrate the potential of this green gel fuel to enhance propulsive efficiency while providing safer handling characteristics for aerospace applications.

1. Introduction

In rocket propulsion, there is a class of propellants known as hypergolic. The term hypergolic refers to the spontaneous ignition that occurs when a fuel and an oxidizer come into contact at room temperature [1]. Hypergolic propellants offer the advantage of reducing ignition failure risks and simplifying the propulsion system compared to non-hypergolic propellants [2].

Hydrazine (N₂H₄) is commonly used as a monopropellant, whereas hypergolic bipropellant systems typically involve fuels such as monomethylhydrazine (MMH) or unsymmetrical dimethylhydrazine (UDMH) in combination with oxidizers like nitrogen tetroxide (NTO, N₂O₄) or mixed oxides of nitrogen (MON) [1]. However, these propellants are carcinogenic, toxic, and corrosive, making them highly hazardous to handle and store. As a result, green propellants have been investigated

as potential alternatives. A green propellant consists of a fuel–oxidizer combination that is less toxic than hydrazine and its derivatives [1,3].

Concentrated hydrogen peroxide (H₂O₂) is a promising candidate for liquid propellant propulsion systems to replace traditional toxic oxidizers such as nitric acid and NTO. At 90% concentration by weight, H₂O₂ exhibits a similar density, lower vapor pressure, and reduced corrosiveness compared to these oxidizers [3]. In general, concentrated H₂O₂ is employed as an oxidizer in combination with alcohols, such as ethanol and butanediol, and with amines, such as monoethanolamine, which are utilized as fuels in bipropellant systems [4,5]. However, liquid propellants present safety and handling challenges. Therefore, gel propellants can be considered a viable alternative [6].

Gel propellant is a type of liquid propellant that has a gelling agent in its composition and presents non-Newtonian behavior [1]. Gelled propellants in aerospace propulsion systems present notable advantages

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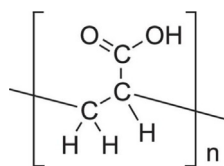


Fig. 1. General structure of Carbopol polymer [18].

in terms of safety and handling compared to liquid propellants. Their stable internal structure enhances storage capability and reduces the risks associated with spillage and leakage, offering a more reliable alternative. In addition, gelled propellants reduce sloshing in tanks, and their shear-thinning behavior allows the flow into the combustion chamber to be precisely controlled or even interrupted when required [1].

The gelling agents commonly used in aerospace propulsion can be classified as organic or inorganic. Inorganic gelling agents consist mainly of fumed silica and also bentonite clays [7,8].

These materials have been widely employed in the formulation of gelled hydrocarbon-based fuels. Recent developments have explored kerosene-based gel formulations [9]. In this study, a kerosene-based gel fuel formulated with fumed nanosilica (SiO_2) as the gelling agent and sodium borohydride (NaBH_4) as a hypergolic additive, combined with hydrogen peroxide as the oxidizer, demonstrated hypergolic ignition with ignition delay times as low as 4–5 ms. These findings highlight the growing interest in gel propulsion systems as viable alternatives for green propulsion technologies.

Organic gelling agents, in contrast to inorganic ones, include a variety of cellulose derivatives, polysaccharides, agarose [10], hydroxypropylcellulose, Carbopol [11–13], polyvinyl alcohol [14], and Thixatrol ST [15]. Carbopol 940, a commercially available form of poly(acrylic acid) derived from acrylic acid ($\text{C}_3\text{H}_4\text{O}_2$), was selected as the gelling agent for the gel propellant formulations. Carbopol 940 (CAS Registry Numbers 9003-01-4 and 76050-42-5) is a high molecular weight carbomer with the chemical formula $(\text{CH}_2-\text{CHCOOH})_n$ (Fig. 1). The estimated molecular weight between crosslinks (M_c) for Carbopol 940 is 104400 g/mol, and it contains between 56% and 68% carboxylic acid (COOH) groups, calculated on a dry basis [16,17]. Carbopol is widely used for cosmetics, pharmaceutical applications, control drug release, emulsification [18,19].

In this context, Carbopol 940 was selected as the gelling agent due to its ability to form stable gel structures at low concentrations, enabling homogeneous systems with suitable consistency for the incorporation of solid particles.

Gelled propellants allow the inclusion of energetic metals and catalytic additives, providing hypergolic ignition for conventional, non-toxic propellant pairs [8]. Metallized gel fuels have been reported in the literature as an effective option to enhance the energy density and calorific value of fuels [1,8,20,21]. Energetic solid additives, such as aluminum (Al) and boron (B), have been investigated to improve combustion characteristics and increase the volumetric energy density [22, 23].

However, the ignition and combustion behavior of gel fuels is influenced not only by the presence of metal particles, but also by their concentration [14].

B.V.S. Jyoti et al. (2017) studied the hypergolicity and ignition delay time of gelled ethanol, formulated with 6% by weight of propyl cellulose as the gelling agent and 1% of a catalyst, either copper chloride hydrate or manganese(II) acetylacetonate. The gels were evaluated in pure form and in separate formulations energized with nano aluminum (Al), boron (B), or carbon (C) particles, using 90% hydrogen peroxide as the oxidizer. In that study, ignition was generally obtained within a few drops of oxidizer addition, as commonly reported for gelled propellants. Gelled ethanol, both in pure and energized

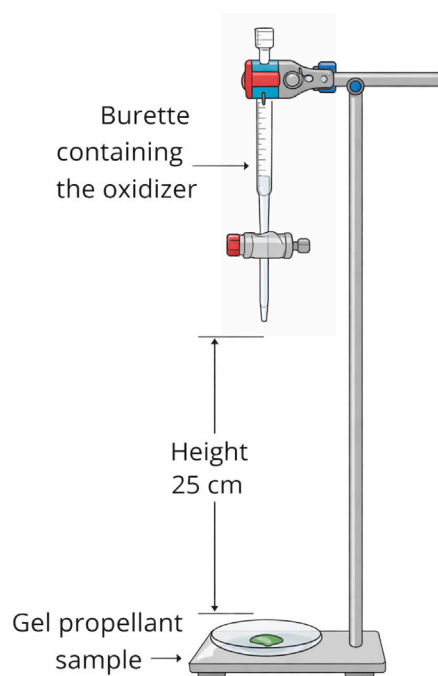


Fig. 2. Experimental apparatus used in the drop test.

forms, exhibited ignition delays comparable to those of existing liquid hypergolic bipropellant systems. The authors also reported that the use of metallic additives improved ignition behavior relative to non-energized gels, indicating the relevance of incorporating energetic particles into ethanol-based hypergolic fuels. The study demonstrated that gelled ethanol propellants offer the advantages of lower toxicity, higher density, and reasonably short ignition delays.

J. Chen et al. (2024) investigated ethanolamine-based gel propellants prepared with 4% agarose as the gelling agent and varying additions of aluminum and carbon particles. The study focused on how carbon powder influences the combustion behavior and rheological characteristics of aluminized gels. Results indicated that adding about 5% carbon improved the combustion efficiency, reduced aluminum particle agglomeration, and contributed to favorable rheological properties that support effective atomization during injection. Furthermore, combustion heat measurements confirmed an improvement in the energetic performance of the gel fuels.

However, research on gelled and metallized gelled propellants remains limited, particularly with regard to performance and efficiency parameters such as specific impulse (I_{sp}), characteristic velocity (c^*), and volumetric specific impulse (ρI_{sp}). Moreover, additional studies are needed to determine the optimal concentrations of metal particles in the fuel formulation in order to maximize energy performance [14]. The performance of a rocket engine is commonly characterized by the specific impulse, in seconds, defined as the thrust-to-propellant weight flow rate at Earth's surface and sea level. The volumetric specific impulse is another relevant parameter, especially when a reduced tank volume is required, as it accounts for both the density of the propellant mixture and its specific impulse. In addition, the characteristic velocity is an efficiency parameter defined at the point where the gas flow becomes sonic at the throat. It reflects the energy potential of the propellant and the efficiency of the injector–combustion chamber system and depends on the properties of the combusted gas at the nozzle inlet [3].

Therefore, this work aims to develop and optimize a green hypergolic gel propellant based on ethanol, employing Carbopol 940 as the gelling agent, copper(II) chloride as the catalyst, and aluminum as the energetic additive, which is less toxic and safer for the environment

than hydrazine. Another objective is to achieve maximum performance and efficiency parameters, such as specific impulse, volumetric specific impulse, and characteristic velocity for aerospace propulsion systems, using NASA CEA software. A distinctive aspect of this work is the application of the design of experiments, which enables the development of predictive models for each response and the optimization of conditions based on statistical analysis.

2. Materials and methods

2.1. Chemicals

Ethanol, used as the fuel, was obtained from Synth. Carbopol, used as the gelling agent, was obtained from Quimisul (SC, Brazil). Copper(II) chloride dihydrate, used as the catalyst, was obtained from Synth. Aluminum, used as the energetic particle, 99% purity, with a maximum particle size of 44 μm , corresponding to a 350 mesh sieve opening, was also obtained from Synth. Hydrogen peroxide (60%), used as the oxidant, was obtained from Solvay Brasil and concentrated to 90% using the propellant-grade peroxide concentration unit available at the Propellants Research Laboratory (LAPEP) of the Aeronautics Institute of Technology.

2.2. Gel propellant preparation

First, the gel was prepared by mixing ethanol with incremental additions of the gelling agent Carbopol until reaching 2.5% (w/w). The mixture was stirred using a magnetic stirrer with a magnetic bar until a homogeneous gel was formed. Approximately 5 g of this gel was then mixed with the catalyst ($\text{CuCl}_2 \cdot 2\text{H}_2\text{O}$) and aluminum particles, when used, using a magnetic stirrer. The quantities of the components were varied according to the applied design of experiments.

Although gel propellants typically exhibit non-Newtonian behavior, including shear-thinning characteristics, rheological measurements (e.g., viscosity and flow behavior) were not performed in the present study. The experimental approach was focused on the identification of formulations that exhibited homogeneous gel formation and maintained their macroscopic consistency during preparation and evaluation. A comprehensive rheological analysis requires dedicated instrumentation and is more directly related to the study of flow behavior and injection system design. Therefore, it is beyond the scope of this work and will be addressed in future studies.

2.3. Drop test

Although the drop test provides a qualitative assessment of hypergolicity, ignition delay time (IDT) is a key parameter for a comprehensive characterization of hypergolic systems. In the present study, the drop test was employed as a screening method to delimit the hypergolic region used in the experimental design. Quantitative evaluation of IDT requires dedicated experimental setups with appropriate temporal resolution, and was therefore not performed in this work.

The drop test was conducted to investigate the hypergolic behavior of the gel propellants and to determine the experimental region in which this behavior occurs. This region corresponds to the studied levels of the factors, which are the catalyst and aluminum concentrations.

Hydrogen peroxide 90% was placed in a burette positioned 25 cm above a watch glass containing 1 g of gel, in order to ensure that the flame produced during ignition did not reach the oxidizer in the burette, thereby maintaining safe operating conditions (Fig. 2). Each drop (0.05 mL) 90 wt% H_2O_2 was released from a burette positioned 25 cm above the fuel. The gel propellant was considered hypergolic when ignition occurred with a maximum of ten drops of oxidizer.

2.4. Experimental region delimitation

Defining the experimental region is essential for the application of the design of experiments, since all generated samples must be analyzable [24]. In this study, the experimental region was delineated through preliminary experimental tests aimed at establishing the lower and upper limits of each formulation component, ensuring hypergolic behavior with 90 wt% hydrogen peroxide while preserving the physicochemical stability required for proper gel formation. Initially, the minimum concentration of the gelling agent (Carbopol) was determined as the lowest concentration capable of producing a stable system in which the solid components remained adequately dispersed, without visible phase separation during preparation and evaluation. Based on this criterion, the gelling agent concentration was fixed at 2.5 wt%.

Subsequently, the maximum and minimum amounts of catalyst and metal particles were defined based on compositions that exhibited homogeneous appearance, characterized by the absence of visible segregation of the components, and consistent hypergolic ignition behavior in triplicate under the same conditions with H_2O_2 (90%). As part of the screening procedure, compositions requiring more than ten drops of oxidizer to achieve ignition were not considered hypergolic.

Once the experimental region was established, all subsequent tests and experimental designs were conducted strictly within these limits, enabling consistent analysis and optimization of the propulsive performance parameters of the developed gel propellant.

2.5. Design of experiments

According to the experimental region delimitation, the maximum and minimum levels were established for the factors under investigation. The type of design of experiments used was a Rotatable Central Composite Design (RCCD), as shown in Fig. 3. RCCD was employed to optimize the specific impulse (I_{sp}), characteristic velocity (c^*), and volumetric specific impulse (ρI_{sp}) of the gel propellant. The gel propellant formulation included Carbopol (gelling agent), ethanol (fuel), copper(II) chloride dihydrate (catalyst), and, when applicable, aluminum particles (energetic additive). Two independent variables (factors) were considered: the concentration (wt%) of copper (II) chloride dihydrate (Cu^{2+}) (x_1), and the concentration (wt%) of aluminum (x_2). The concentration of the gel was fixed to 2.5% Carbopol in ethanol. The RCCD consisted of a full 2^2 factorial design, four axial points (± 1.41), and center points, resulting in 11 total experiments. The center point (point 9 in Fig. 3) was evaluated in triplicate to allow the assessment of experimental variability, particularly associated with the density measurements used in the calculation of volumetric specific impulse. Axial levels (± 1.41) were set according to the experimental region boundaries, while other levels followed RCCD structure. The RCCD provides a second-order predictive model for the studied responses (Eq. (1)).

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_{11} x_1^2 + \beta_{22} x_2^2 + \beta_{12} x_1 x_2 \quad (1)$$

Where y indicates the response (I_{sp} , c^* , and ρI_{sp}); x_1 and x_2 are the factors including catalyst (x_1) and aluminum concentration (x_2), β_0 is the constant coefficient of the model; $\beta_1 x_1$ and $\beta_2 x_2$ are the regression coefficients of the single effects of factors; $\beta_{11} x_1^2$ and $\beta_{22} x_2^2$ are the quadratic effects coefficients; $\beta_{12} x_1 x_2$ is the interaction coefficient.

2.5.1. Analyses of variance (ANOVA)

Analyses of variance (ANOVA) were applied to evaluate the predictive models and to verify their adequacy for the studied responses. ANOVA verifies the significance of the model and the lack of fit, while the behavior of the standardized residuals was assessed separately through diagnostic plots. The assumption of normality was verified using the normal probability plot of the residuals. If the residuals follow a normal distribution, the points on the plot will approximately form a straight line, and ANOVA is considered robust to the normality

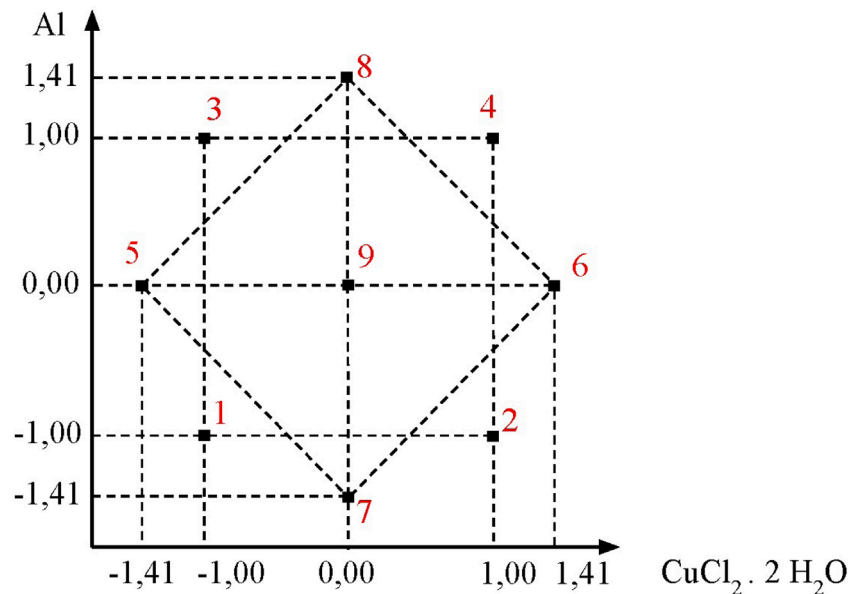


Fig. 3. Schematic representation of RCCD.

assumption. In the statistical analysis, we assume that the errors are normally and independently distributed random variances. To avoid any violation of the independence assumption, the experiments were properly randomized. A plot of residuals versus fitted values was analyzed to infer whether the model is appropriate and the assumptions are satisfied. The plot should not exhibit an apparent pattern [24].

2.6. Density

Density is a parameter that affects propellant performance. High-density propellants can be stored in smaller tanks. By contrast, low-density propellants require larger tanks and, as a result, increase the overall mass of the propulsion system [3]. The density was measured using a metal pycnometer with a 25 mL capacity. First, the pycnometer was calibrated with distilled water at 23 °C. Then, it was weighed with the gel propellant sample at the same temperature. The density was calculated according to Eq. (2).

$$\rho = \frac{N - M}{V} \quad (2)$$

Where ρ is the density of the gel propellant (g/cm³), M is the mass of the empty pycnometer (g), N is the mass of the pycnometer filled with approximately 25 mL of gel propellant sample (g), and V is the volume of the pycnometer (cm³). The density was then calculated using Eq. (2) [25,26]. The final results were expressed in kg/m³.

2.7. Performance and efficiency parameters

Volumetric specific impulse, ρI_{sp} , is defined as the product of the bulk propellant density and its specific impulse [27,28]. This parameter can be used to determine the total volume of tankage required. High values of both propellant density and volumetric specific impulse are desirable to reduce the volume and weight of the tanks and pumps in rocket engines [3].

To calculate ρI_{sp} , it is necessary to determine the theoretical specific impulse (I_{sp}), which represents the total impulse per unit weight of propellant. This parameter measures the efficiency of a rocket propulsion system in which higher values of I_{sp} indicate better performance [1].

Another required parameter is the bulk propellant density (ρ_p), calculated according to Eq. (3). This density expresses the ratio between the total mass of the propellant and the total volume it occupies

and is written in terms of the proportions of fuel and oxidizer. The performance of bipropellant systems can also be characterized by the mixture ratio (O/F), defined as the mass flow rate ratio of oxidizer to fuel [27].

$$\rho_p = \frac{1 + \frac{O}{F}}{\frac{1}{\rho_f} + \frac{O/F}{\rho_{ox}}} \quad (3)$$

Where ρ_p is the density of the propellant mixture (kg/m³), ρ_f is the fuel density (kg/m³), ρ_{ox} is the oxidizer density (kg/m³), and O/F is the oxidizer-to-fuel mass ratio. The oxidizer density used, corresponding to 90 wt% H₂O₂, was 1388.9 kg/m³.

The volumetric specific impulse is then obtained using Eq. (4) [28]:

$$\rho I_{sp} = I_{sp} \times \rho_p \quad (4)$$

Another parameter used in this study was the characteristic velocity (c^*), which relates to the combustion efficiency and is independent of nozzle characteristics. This parameter is generally used to compare the relative performance of different chemical rocket propulsion system designs and propellant combinations [1].

2.8. NASA's chemical equilibrium with applications simulation

The theoretical specific impulse (I_{sp}) and characteristic velocity (c^*) were estimated using the Chemical Equilibrium with Applications (CEA) code developed by NASA [29]. Then, the O/F ratio was selected according to the highest values of these responses. This software calculates chemical equilibrium compositions and provides the corresponding thermodynamic and transport properties of the system. Performance analyses were performed under equilibrium conditions, with a chamber pressure of 5 bar and an exit pressure of 1.02 bar, resulting in a pressure ratio of 4.9 [30]. Simulations were performed for different oxidizer-to-fuel mass ratios, based on the optimized RCCD formulation. The initial temperature was set to 298 K, with the physical states of the reactants indicated as L for liquid and S for solid. Table 1 lists the standard enthalpy of formation for all species. Since the value for copper(II) chloride dihydrate was not available in the NASA CEA database, a literature source was used. For the remaining substances, data from the NASA CEA library were employed.

Table 1
Standard enthalpy of formation (kJ/mol) at T = 298 K for the species used in NASA CEA.

Species Name	Formula	MW (g/mol)	ΔH_f° (kJ/mol)	Ref.
Aluminum (s)	Al	27.0	0	McBride et al. [31]
Carbopol (L)	$C_3H_6O_2$	76.06	−383.6	National Institute of Standards and Technology (NIST) [32]
Cu^{2+} (S)	$CuCl_2 \cdot 2H_2O$	241.60	−1217	Polyachenok et al. [33]
Ethanol (L)	C_2H_6O	46.07	−277.51	McBride et al. [31]
Hydrogen peroxide (L)	H_2O_2	34.01	−187.78	McBride et al. [31]
Water (L)	H_2O	18.02	−286.30	McBride et al. [31]

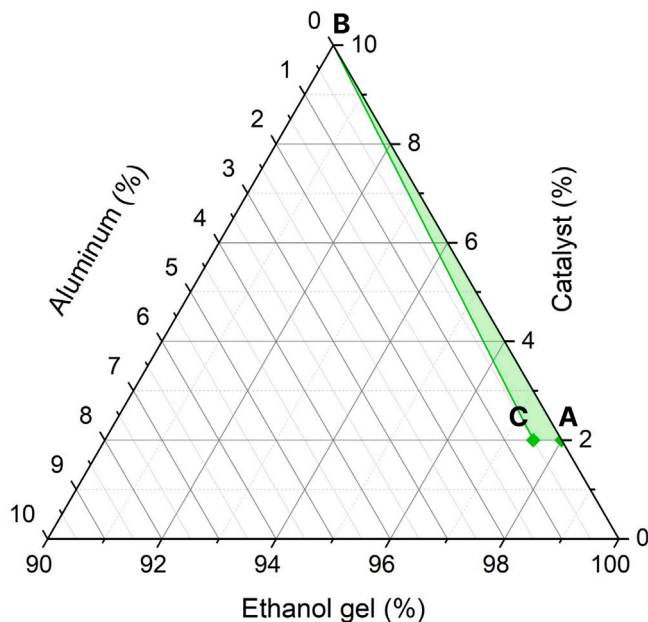


Fig. 4. Experimental region delimitation.

3. Results and discussion

3.1. Experimental region delimitation

The experimental region was evaluated to determine the minimum and maximum catalyst and aluminum concentrations that exhibited hypergolicity. Fig. 4 illustrates this region, with Points A, B and C marking its boundaries. The points A and B represent the minimum and maximum catalyst concentrations (%), respectively. Point B also corresponds to the formulation without aluminum particles (0%), since aluminum is not required to trigger the hypergolic reaction. Point C corresponds to the formulation with the highest aluminum content, achieved when the catalyst is at its minimum. Aluminum contents above 0.5% are not recommended, as excessive aluminum concentration makes the mixture excessively dense and rigid, resulting in the loss of its gel-like consistency and a transition to solid-like behavior due to high solid loading. Once the experimental region was established, all subsequent tests were conducted within these limits, enabling the optimization of the propulsive parameters and the energy efficiency of the developed gel propellant.

3.2. Optimization of propulsive performance using RCCD models

The main objective of this study was to investigate the performance and efficiency parameters, namely I_{sp} , c^* , and ρI_{sp} , of the formulated gel in a bipropellant system. The selected levels of both factors were determined by the experimental region delimitation. The axial points (± 1.41) for $CuCl_2$ were 2 and 10% and 0% to 0.5% for Al. The center point (0) for both $CuCl_2$ and Al was calculated as the midpoint of their respective ranges, corresponding to 6% and 0.25%, respectively. This

Table 2
Experimental results for density and ρ_p .

Experiment	Density (kg/m ³)	ρ_p (kg/m ³)
1	821.4	1220
2	854.1	1234
3	822.7	1221
4	854.9	1235
5	814.2	1217
6	862.0	1238
7	836.8	1227
8	839.6	1228
9	838.2	1228
10	840.2	1228
11	839.4	1228

Table 3
Experimental design matrix (coded levels) and corresponding results for I_{sp} (s), c^* (m/s), and ρI_{sp} .

Experiments	Coded factors		Uncoded factors		Responses		
	x_1	x_2	$CuCl_2$ (%)	Al (%)	I_{sp} (s)	c^* (m/s)	ρI_{sp} (kg.s/m ³)
1	−1.00	−1.00	3.2	0.07	172.53	1566	210,537
2	+1.00	−1.00	8.8	0.07	171.69	1559	211,920
3	−1.00	+1.00	3.2	0.43	172.60	1567	210,719
4	+1.00	+1.00	8.8	0.43	171.74	1559	212,041
5	−1.41	0.00	2.0	0.25	172.70	1568	210,188
6	+1.41	0.00	10.0	0.25	171.48	1557	212,221
7	0.00	−1.41	6.0	0.00	172.15	1563	211,224
8	0.00	+1.41	6.0	0.50	172.23	1564	211,533
9	0.00	0.00	6.0	0.25	172.19	1563	211,377
10	0.00	0.00	6.0	0.25	172.19	1563	211,528
11	0.00	0.00	6.0	0.25	172.19	1563	211,466

condition was evaluated in triplicate (runs 9, 10, and 11). The factorial points (± 1) were obtained by interpolating values between the axial points. The combinations of these intermediate points, located between the axial and center levels, enable the assessment of both the main and interaction effects of the factors on the responses.

Density (kg/m³) and ρ_p (kg.s/m³) results are shown in Table 2. The density values were determined experimentally, while the bulk propellant density was calculated according to Eq. (3). These parameters were subsequently combined with the specific impulse, obtained from NASA CEA simulations, to determine the volumetric specific impulse. This parameter therefore integrates experimentally measured data with results derived from deterministic simulations.

Table 3 presents the factor levels employed and the corresponding results for I_{sp} (s) and c^* (m/s), obtained using NASA CEA simulations, while ρI_{sp} was calculated according to the density results.

Eqs. (5)–(7), with the simplified terms, are presented for I_{sp} , c^* , and ρI_{sp} , respectively. These equations represent predictive models derived from the statistical analysis, enabling the estimation of the propulsive responses as a function of the gel formulation, even for compositions not experimentally evaluated within the studied region.

$$I_{sp} = 173 - 0.07388x_1 + 0.2249x_2 - 0.006250x_1^2 - 0.01000x_1x_2 \quad (5)$$

$$c^* = 1569 - 0.5505x_1 + 2.136x_2 - 0.06820x_1^2 - 0.1500x_1x_2 \quad (6)$$

$$\rho \cdot I_{sp} = 209307 + 415.3x_1 + 523x_2 - 14.06x_1^2 \quad (7)$$

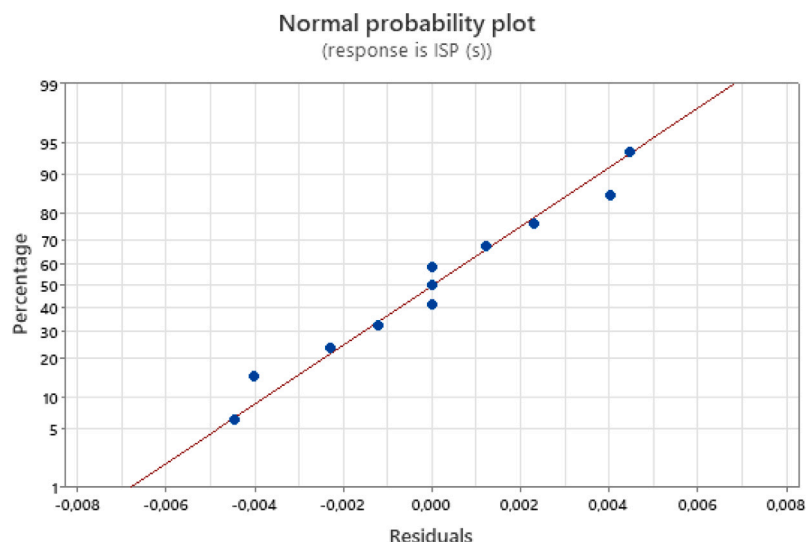


Fig. 5. Normal probability plot for the response I_{sp} (s)

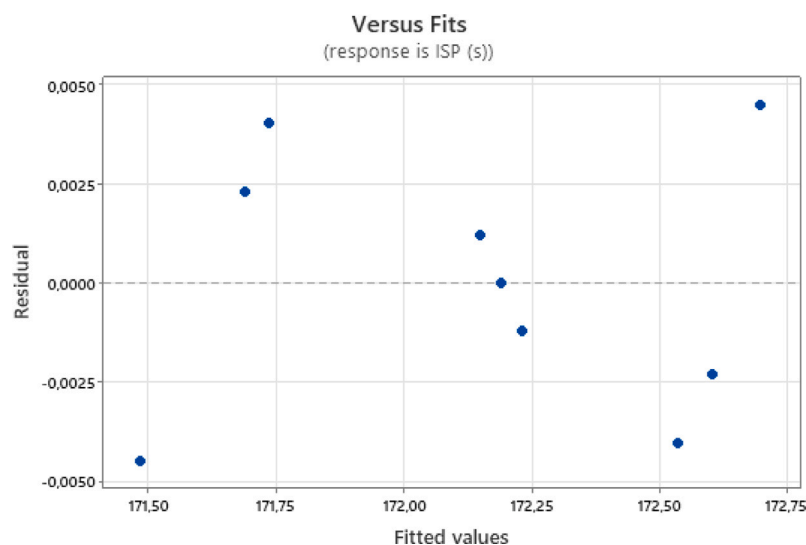


Fig. 6. Residuals versus fitted values plot for the response I_{sp} (s)

3.3. ANOVA

The statistical analyses were performed using Minitab software. Before applying ANOVA, the normal probability plot (Fig. 5) and the residuals-versus-fitted-values plot (Fig. 6) were examined to verify the normality assumption and assess the adequacy of the model. Only the plots for the I_{sp} response are shown here to reduce the number of figures, while the corresponding plots for c^* and ρI_{sp} are provided in Appendix.

According to Figs. 5 and 6, the results indicate that the normality assumption was satisfied and the residuals were structureless.

Tables 4 to 6 show the ANOVA results for the responses I_{sp} , c^* , and ρI_{sp} , respectively. All statistical tests were analyzed using 95% confidence level.

The F-test was applied to evaluate the significance of the model. The calculated F-value (F_{Calc}) was obtained from the ratio between the mean square (MS) of the regression (R) and the mean square of the residuals (r). For the model to be considered significant, F_{Calc} must be greater than the tabulated F-value (F_{Tab}) at a 95% confidence level.

The adequacy of the model was also analyzed using the F-test, by calculating the ratio between the mean square of the lack of fit (LOF)

and the mean square of the pure error (PE). In this case, F_{Calc} should be smaller than the tabulated F-value (F_{Tab}).

The comparison of the calculated F-value (F_{Calc}) to the tabulated F-value (F_{Tab}), at a 95% confidence level, shows that the F_{Calc} values for the responses I_{sp} (17282), c^* (5402), and ρI_{sp} (222) are greater than F_{Tab} (5.05), indicating that the models are significant. The significance of each term in the regression model for the responses was evaluated using a Pareto chart.

For both responses I_{sp} and c^* , the quadratic term x_2^2 (aluminum percentage) was not significant and was eliminated from the equations. For ρI_{sp} , both the quadratic term x_2^2 and the interaction term $x_1 x_2$ (catalyst and aluminum percentage) were excluded from the corresponding model equation.

The adequacy of the models for I_{sp} and c^* showed F_{Calc} equal to zero due to the absence of error in the simulation results. For the response ρI_{sp} , F_{Calc} was lower (0.36) than F_{Tab} (19.16), indicating that the model is adequate.

In addition, the results showed that the models can reliably predict the studied responses reflecting by both R^2 and $R^2_{adjusted}$ of the models which were higher than 99%.

The Pareto charts for each response (Fig. 7) were analyzed to identify the significant terms in the equations and to assess the magnitude

Table 4Analysis of variance (ANOVA) for the I_{sp} model.

Source of Variation	DF	SS	MS	F _{calc}	F _{tab}
Regression (R)	4	1.48897	0.29779	17282	4.53
Residual (r)	6	0.00009	0.00002	–	–
Total	10	1.48905	–	–	–

Explained variation (R^2 , %): 99.99.Adjusted maximum explained variation (R^2_{adjusted} , %): 100.**Table 5**Analysis of variance (ANOVA) for the c^* model.

Source of Variation	DF	SS	MS	F _{calc}	F _{tab}
Regression (R)	4	128.825	32.206	8030	4.53
Residual (r)	6	0.024	0.004	–	–
Total	10	128.849	–	–	–

Explained variation (R^2 , %): 99.98.Adjusted maximum explained variation (R^2_{adjusted} , %): 100.**Table 6**Analysis of variance (ANOVA) for the ρI_{sp} model.

Source of Variation	DF	SS	MS	F _{calc}	F _{tab}
Regression (R)	3	4038810	1346270	363	4.35
Residual (r)	7	25982	3712	–	–
Lack of fit (LOF)	5	14460	2892	0.50	19.30
Pure error (PE)	2	11522	5761	–	–
Total	10	4064792	–	–	–

Explained variation (R^2 , %): 99.36.Adjusted maximum explained variation (R^2_{adjusted} , %): 99.72.**Table 7**Optimized values of I_{sp} and c^* using NASA-CEA.

Solution	Catalyst (%)	Aluminum (%)	I_{sp} (s)	c^* (m/s)
1	2.00	0.50	172.8	1568

of their effects. These charts display the absolute values of the standardized effects, ordered from largest to smallest, with a reference line indicating the threshold for statistical significance. The factors were statistically analyzed at the 95% confidence level using the current model terms.

For all responses, the terms A (principal term of catalyst concentration), AA (quadratic term of catalyst concentration), and B (principal term of aluminum concentration) were the most significant, in order of relevance. For I_{sp} and c^* , the interaction term AB was also significant. For ρI_{sp} , however, this interaction was not significant and it was removed from the model. The quadratic term of B (aluminum concentration) was also excluded from all predictive models. Fig. 8 represents the response surfaces and contour plots for (a) I_{sp} , (b) c^* , and (c) ρI_{sp} , respectively.

Based on the results for I_{sp} , c^* , and ρI_{sp} , both catalyst and aluminum concentrations (%) were found to simultaneously influence the responses. The highest response for I_{sp} and c^* values occurred at intermediate levels of catalyst and higher levels of aluminum using H_2O_2 (90%) as the oxidant.

For the response ρI_{sp} , the catalyst concentration had a greater impact than the aluminum concentration. Higher catalyst content resulted in higher density, even with reduced or absent aluminum.

The optimized conditions for the factors, as predicted by the models for I_{sp} and c^* , are presented in Table 7, whereas the optimized conditions for ρI_{sp} are shown in Table 8.

According to the values obtained from the optimization, a new simulation was performed using NASA CEA to compare the estimated values from the predictive models with those obtained from the simulation. The results from the simulation were very close to the predicted values,

Table 8Optimized values of ρI_{sp} (kg.s/m³).

Solution	Catalyst (%)	Aluminum (%)	ρI_{sp}
1	10.0	0.50	212315

with $I_{sp} = 172.8$ s and $c^* = 1568$ m/s at O/F 4.0. This indicates that the predictive models were efficient and can be reliably applied in propellant development. In addition, these values are higher than those of the optimized bipropellant fuel based on 61 wt% monoethanolamine, 30.1 wt% ethanol, and 8.9 wt% copper nitrate developed by Maschio et al. [4], which, under the same conditions, showed an I_{sp} of 160.7 s and a c^* of 1456 m/s at an O/F ratio of 2.6. Furthermore, the optimized bipropellant fuel consisting of 60.0 wt% monoethanolamine, 36.0 wt% n-butanol, and 4.0 wt% copper nitrate trihydrate, developed by Mendoza et al. [5], under the same conditions, exhibited an I_{sp} of 168.8 s and a c^* of 1533 m/s at an O/F ratio of 4.3. This comparison shows that the gel propellant would increase the I_{sp} and c^* parameters and also shifts the optimal O/F to a higher value. Working at a higher O/F increases the combustion temperature and promotes a more complete burn of the propellant, which can improve performance up to the mixture's optimal point.

In this context, Maschio et al. (2018) investigated approximately 20 catalyst formulations for ethanol–hydrogen peroxide hypergolic systems, including copper(II) chloride and nickel chloride, which may lead to the formation of trace chlorine-containing species. Similarly, Mendoza et al. (2025) evaluated hypergolic combinations of n-butanol and monoethanolamine using copper nitrate-based catalysts, focusing on ignition performance and reaction behavior. These studies highlight the importance of catalyst selection in hypergolic systems. In the present work, the selected catalyst contains chlorine, which may lead to the formation of small amounts of hydrogen chloride (HCl) in the exhaust gases. Equilibrium calculations indicate that HCl formation is limited to trace levels (0.096 mol%). Although HCl is a corrosive and toxic species, its concentration in this system is very low. Furthermore, this result should be interpreted within the broader context of green propulsion. Conventional hypergolic propellants, such as hydrazine and nitrogen tetroxide, are highly toxic and carcinogenic. In contrast, the formulation proposed in this work significantly reduces overall toxicity, with only trace formation of potentially harmful byproducts. Nevertheless, the presence of chlorine-containing species may represent a limitation for certain applications, and this aspect should be further investigated in future studies, particularly regarding material compatibility and long-term operational effects.

In addition to the chemical aspects related to catalyst composition, the presence of metallic additives also plays an important role in propulsion performance. However, the fuel used in this study contains an initial aluminum fraction of only 0.5 wt%, which is tens to hundreds of times lower than the typical concentrations (> 15 wt%) employed in solid rocket motors [1]. Regarding the exhaust gases, it is well known that the presence of condensed products significantly affects two-phase flow in nozzles and, consequently, engine performance. The molar fractions of Al_2O_3 in the exhaust products were determined from chemical equilibrium calculations using the NASA CEA code. The values obtained were 0.039% for O/F = 4 and 0.049% for O/F = 2.6. Hill and Peterson [34] presented a simplified model to estimate the specific impulse loss associated with the ejection of solid particles through the nozzle, considering only the mass fraction of condensed products (without accounting for particle size effects). By applying this model to the results obtained from the CEA calculations for the analyzed propellant, this indicates that, since the mass fraction of condensed species is much lower than that of gaseous products, the estimated theoretical I_{sp} losses are approximately 0.19% for O/F = 4 and 0.26% for O/F = 2.6. For comparison, Mueller and Turns [35] reported that gel propellants containing 60 wt% aluminum exhibited

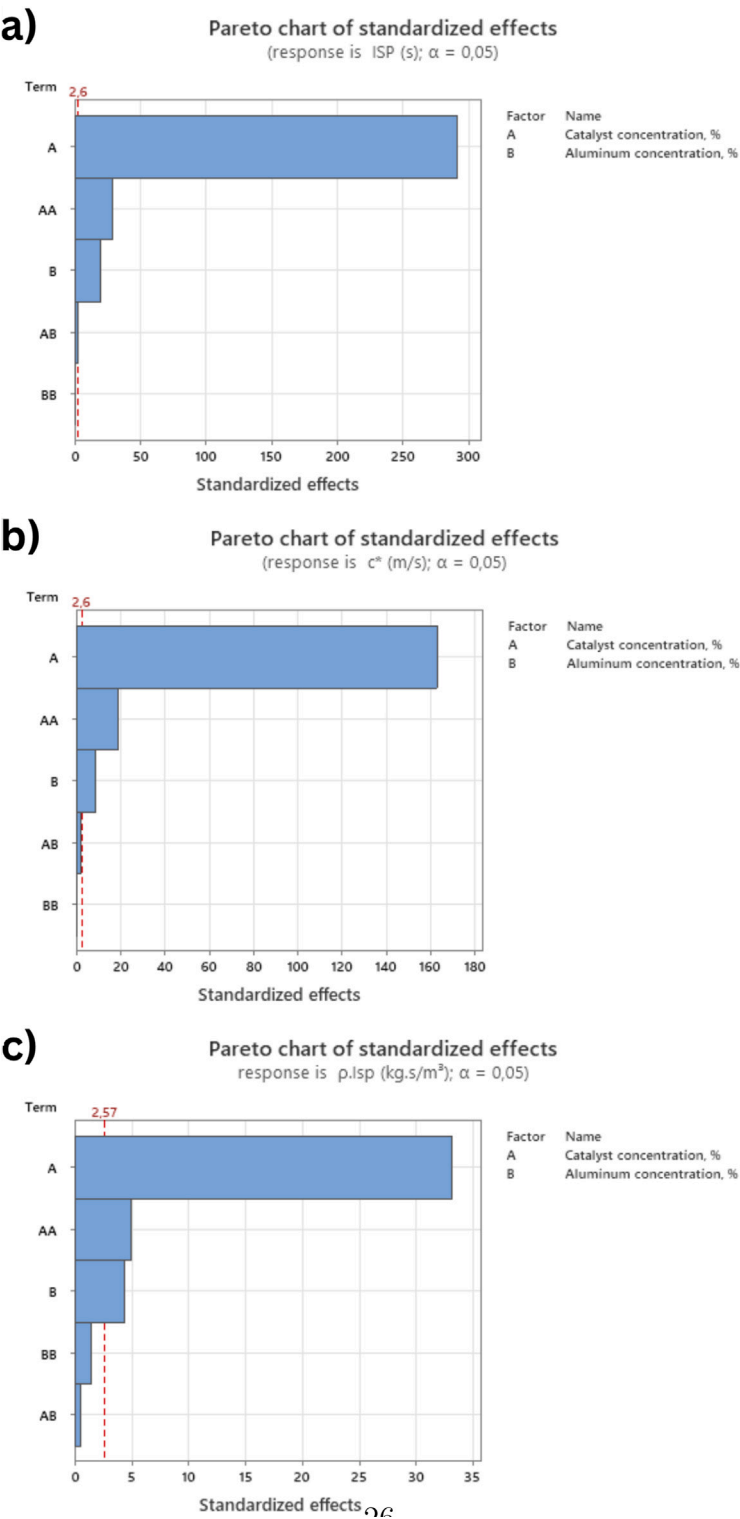


Fig. 7. Pareto chart for the response (a) I_{sp} , (b) c^* and (c) ρI_{sp} .

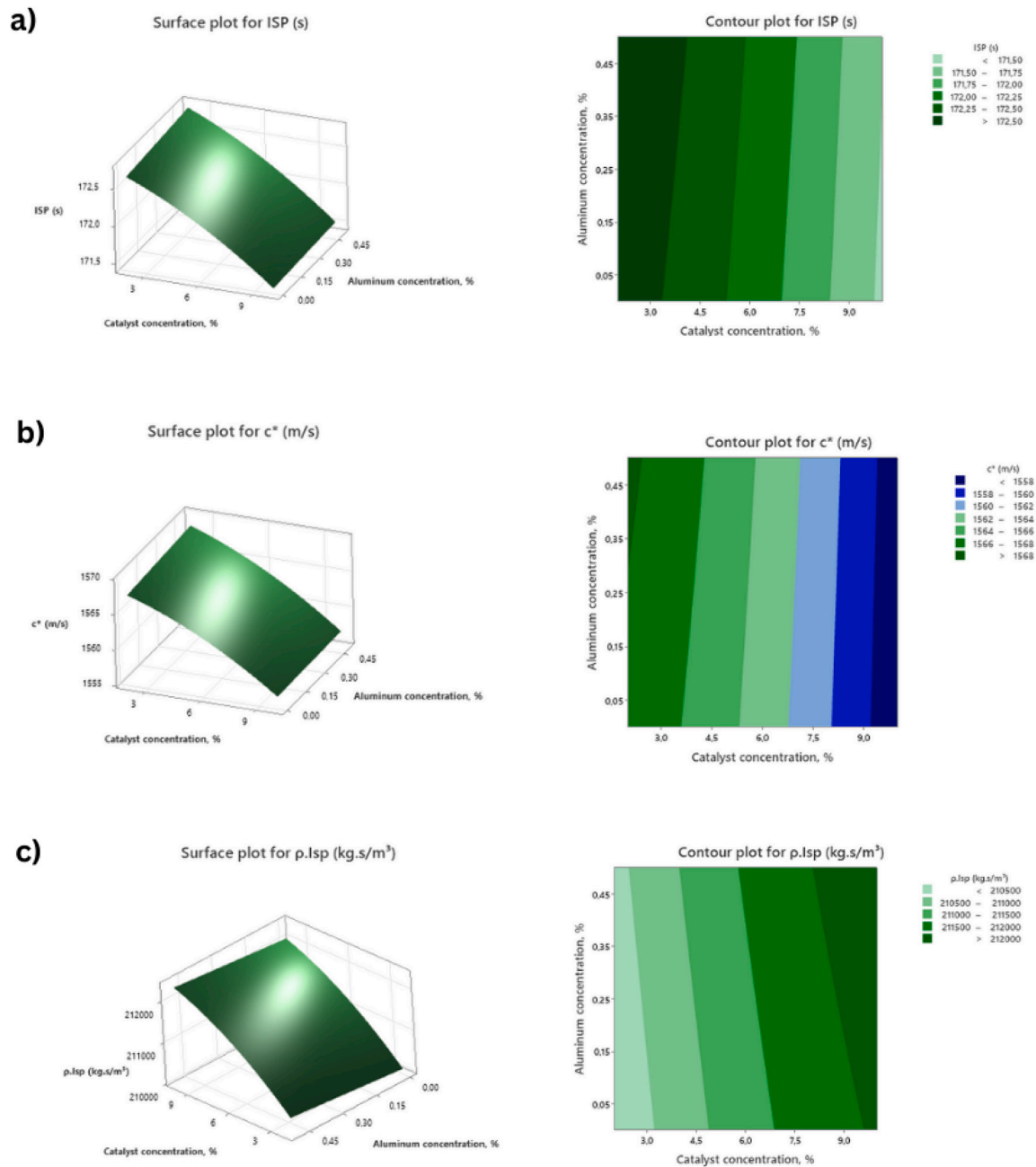


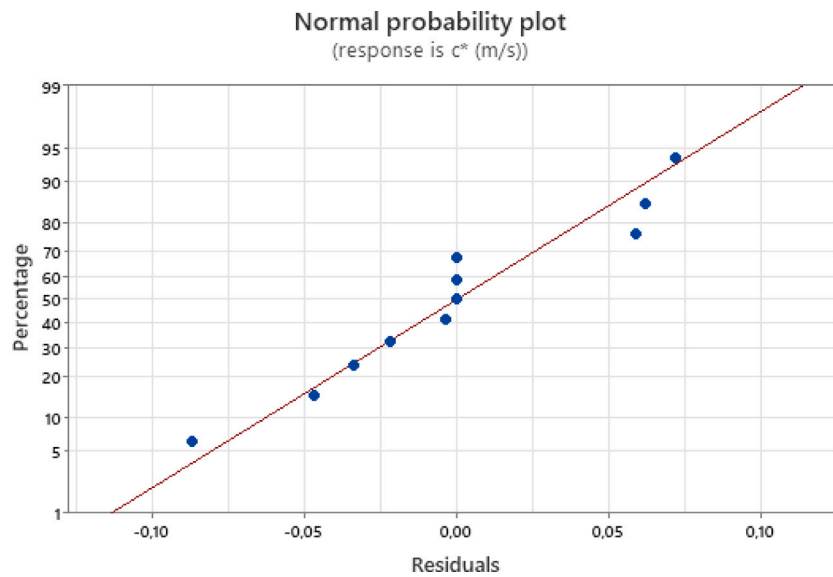
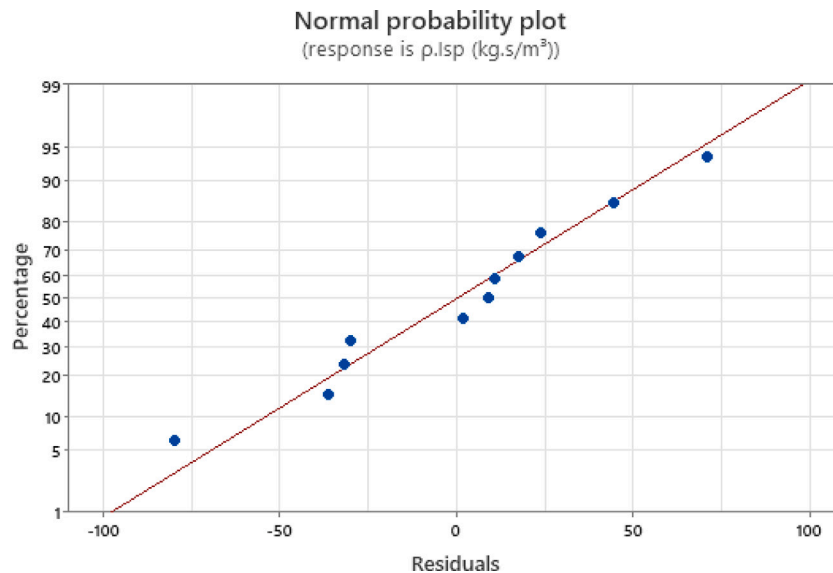
Fig. 8. Surface plot and contour plots for response (a) I_{sp} , (b) c^* and (c) ρI_{sp} .

I_{sp} losses on the order of 5%. Therefore, considering the low mass fraction of condensed products in the present case and the absence of experimental data or particle size distribution, such corrections were neglected. Nevertheless, it should be noted that the influence of condensed products tends to become significant in formulations with higher aluminum content.

4. Conclusion

In conclusion, this work demonstrated the optimization of a hypergolic green gel propellant based on ethanol, Carbolpol, copper(II) chloride, and aluminum particles, using 90% hydrogen peroxide as the oxidizer. By using a Rotatable Central Composite Design (RCCD) methodology, quadratic models were obtained for specific impulse (I_{sp}), characteristic velocity (c^*), and volumetric specific impulse, allowing optimization of these responses. The optimization identified

the composition of 97.5% gel propellant based on ethanol and 2.5% Carbolpol, with 2% copper(II) chloride dihydrate ($\text{CuCl}_2 \cdot 2\text{H}_2\text{O}$) and 0.5% aluminum particles, as yielding the highest values for I_{sp} (172.8 s) and c^* (1568 m/s) at an O/F ratio of 4.0. These values are higher when compared with bipropellant formulations based on MEA, ethanol, and catalyst, as well as MEA, n-butanol, and catalyst, which yielded I_{sp} of 160.7 s and c^* of 1456 m/s at an O/F ratio of 2.6 [4], and I_{sp} of 168.8 s and c^* of 1533 m/s at an O/F ratio of 4.3 [5], respectively. The composition for the optimization of ρI_{sp} was an 89.5% gel propellant based on ethanol and 2.5% Carbolpol, with 10% $\text{CuCl}_2 \cdot 2\text{H}_2\text{O}$ and 0.5% aluminum particles, which yielded 212315 kg.s/m³ for volumetric specific impulse. Optimizing the ρI_{sp} of the gel propellant can provide a competitive advantage for a specific chamber-and-nozzle configuration. Furthermore, the lower toxicity compared to hydrazine-based propellants of this gelled propellant, combined with hypergolic ignition obtained through gelation and particle suspension, highlights its potential for space applications as a viable alternative to hydrazine. Future

Fig. A.9. Normal probability plot for the response c^* (m/s)Fig. A.10. Normal probability plot for the response ρ_{Isp} (kg s/m³)

work will include detailed ignition delay time (IDT) characterization, rheological analysis, and combustion tests to further evaluate ignition dynamics, flow behavior, and practical implementation aspects of the proposed formulations. In addition, long-term storage stability, aging effects, and compatibility with injection systems will be investigated, as these factors are critical for the practical application of gel propellants in propulsion systems.

CRediT authorship contribution statement

Renata Fraga Cardoso: Writing – review & editing, Writing – original draft, Supervision, Project administration, Methodology, Investigation, Formal analysis, Data curation. **Emilly Stefane de Oliveira Fernandes:** Writing – original draft, Validation, Investigation, Formal analysis. **Victor Rabelo Rodrigues:** Writing – review & editing. **Leonardo Henrique Gouvea:** Writing – review & editing, Supervision. **Luís Gustavo Ferroni Pereira:** Writing – review & editing, Validation, Supervision, Resources, Project administration, Methodology, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix. Statistical plots

Normal probability plots and residuals versus fitted values plots for the responses c^* (m/s) and ρ_{Isp} (kg.s/m³) (see Figs. A.9–A.12).

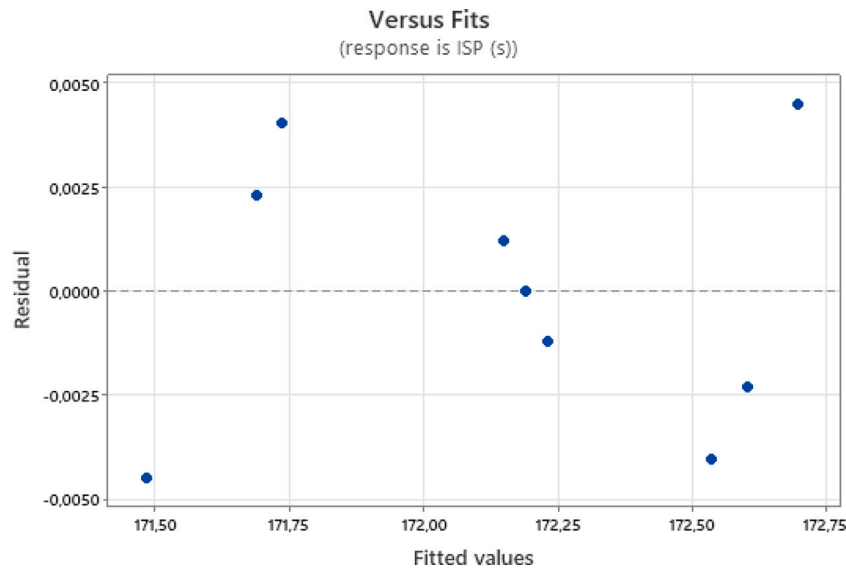


Fig. A.11. Residuals versus fitted values plot for the response c^* (m/s)

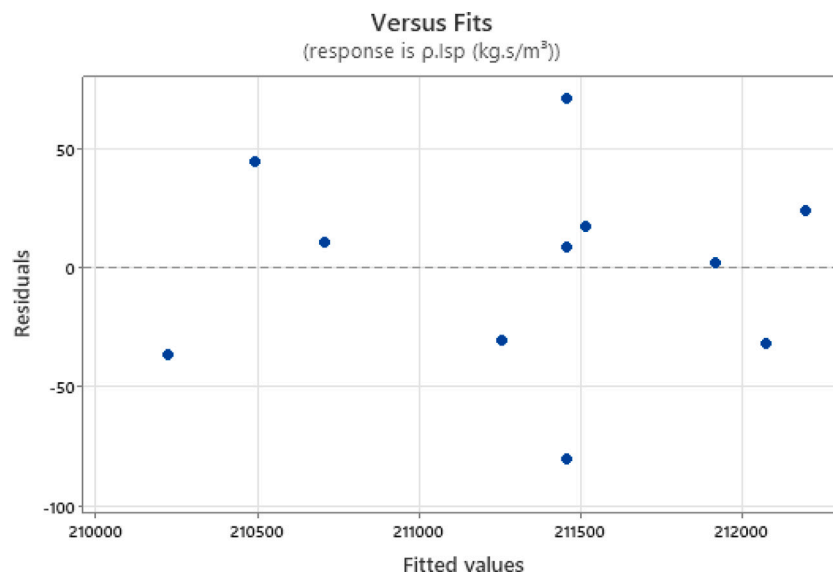


Fig. A.12. Residuals versus fitted values plot for the response ρI_{sp} (kg s/m³)

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